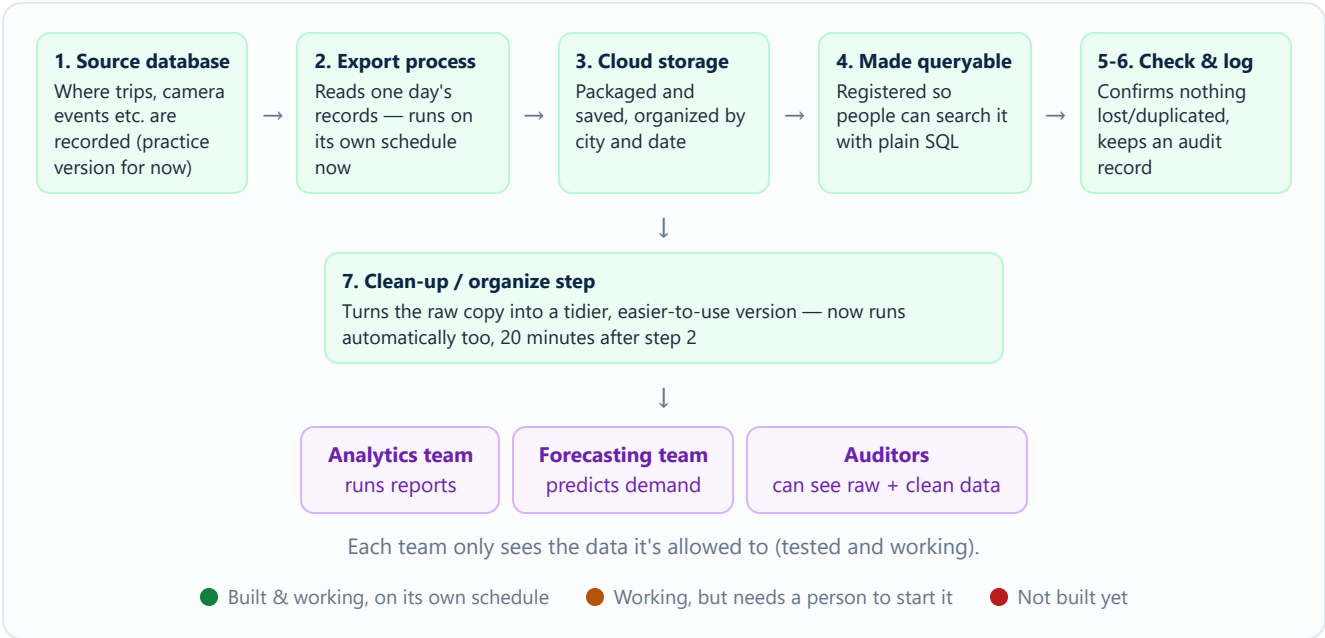


How Data Moves Through the System

A plain-language walkthrough of the flow, from source database to people using it — now running on its own each night



What's automatic today vs. what's not

Step	Status	What that means
1. Data exists in the source database	Working	Real for testing purposes; the actual production database isn't connected yet
2. Export process runs	Automatic	Runs on its own every night — confirmed by watching it fire without anyone starting it
3-6. Storage, cataloging, checking, logging	Working	Fully automatic, no manual work needed
7. Clean-up / organize step	Automatic	Now also runs on its own every night, 20 minutes after step 2 — people no longer only see raw data
8. Teams querying the data	Working	The access rules (who can see what) are built and tested
Catching up on older data	Partly done	A "process many days at once" mode exists and has been run for real over a multi-week range; hasn't been run over the full available history yet

Where to actually see each step happening

All in the AWS Console, region **ap-south-1**, account **690293068614**.

Step	Where to look
1. Source database	RDS → Databases → <code>mtsai-api-sim-test</code>
2. Export process — the schedule itself	EventBridge → Scheduler → <code>mtsai-datalake-test-nightly-export</code>
2. Export process — while it's running	ECS → Clusters → <code>mtsai-datalake-test</code> → Tasks tab (only visible for about an hour after it finishes)

2. Export process — after the fact (reliable)	CloudWatch → Log groups → <code>/mtsai-datalake/test/export</code> → streams starting with <code>export/</code>
3. Cloud storage	S3 → <code>mtsai-datalake-test-690293068614-ap-south-1</code> → <code>raw/</code> folder
4. Made queryable	Athena → Query editor → database <code>test_raw</code> → <code>SELECT * FROM trip_events LIMIT 10</code>
5-6. Check & log — the evidence trail	Same S3 bucket → <code>export-manifests/</code> folder (one file per run)
5-6. Check & log — the logs	Same CloudWatch log group as step 2
7. Clean-up / organize — the schedule	EventBridge → Scheduler → <code>mtsai-datalake-test-nightly-curate</code>
7. Clean-up / organize — the result	S3 → <code>curated/</code> folder, or Athena → database <code>test_curated</code>
7. Clean-up / organize — the logs	Same log group, streams starting with <code>curate/</code>
Costs and alerts (not pictured above)	CloudWatch → Dashboards → <code>mtsai-datalake-test-cost</code>

Generated from the current state of the `mtsai-datalake` project, 21 September 2026. Reflects what's actually built and tested, not the eventual full design. Companion to `MTSAi-Data-Lake-Build-Report.pdf` (technical detail).